Grade 12 — Life Sciences Lebanese curriculum	**Test Preparation 2** Chapter 1 — Energy	Duration: 1 h 30 min Total: 20 points

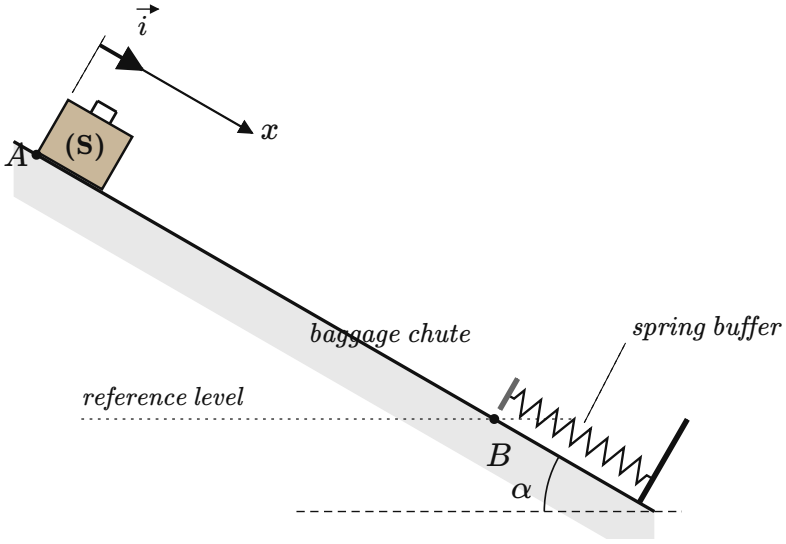
This test is formed of two obligatory exercises. The use of a non-programmable calculator is allowed.

Take, in all exercises: $g = 10\ \text{m/s}^2$.

Exercise 1 — The baggage chute (10 points)

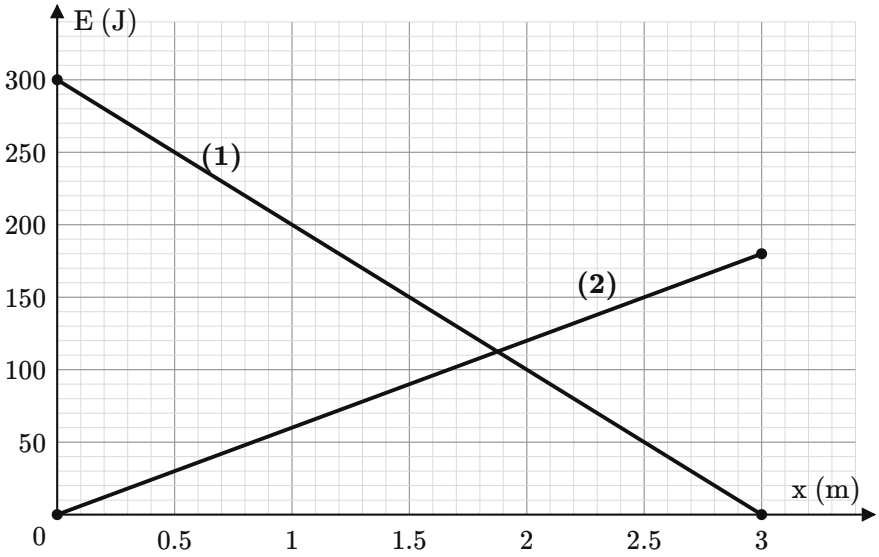
At an airport, suitcases slide down an inclined chute towards the reclaim area. At the lower end of the chute, a spring buffer stops them (Doc. 1). The chute exerts on a suitcase a friction force $\vec{f}$ of constant magnitude, opposite to its velocity.

(not to scale)



Doc. 1

Mass of the suitcase (S): $m = 20$ kg; (S) is considered as a particle.
The chute is inclined by the angle α with the horizontal: $\sin\alpha = 0.5$; $AB = 3.0$ m.
Spring buffer: negligible mass, stiffness $k = 8.0 \times 10^2$ N/m, axis along the chute; its free end is at B when it has its natural length.
At $t_0 = 0$, (S) is released without initial velocity at A. Its position is located by its abscissa x on the axis $(A, \vec{i}\,)$ directed down the chute; v is the algebraic value of its velocity.
The horizontal plane through B is the reference level for gravitational potential energy.
Doc. 2 represents, in terms of x, the kinetic energy and the gravitational potential energy of the system [(S), Earth] between A and B.



Doc. 2

Part 1 — Sliding down the chute (from A to B)

1.1 Express the gravitational potential energy of the system [(S), Earth] in terms of m, g, α, AB and x. Identify, with justification, the two curves of Doc. 2.

1.2 Name the external forces acting on (S) between A and B, and give the expression of the work done by each of them between A and a position of abscissa x.

1.3 Applying the kinetic energy theorem between A and the position of abscissa x, show that $KE = (mg\sin\alpha - f)\,x$. Deduce, using Doc. 2, the value of f.

1.4 Express the mechanical energy of the system [(S), Earth] in terms of m, g, α, AB, x and v. By deriving the relation between this mechanical energy and the work of $\vec{f}$ with respect to time, determine the acceleration a_1 of (S). Specify the nature of its motion.

1.5 Determine the instant t_B at which (S) reaches B and its speed v_B there. Verify v_B using Doc. 2.

1.6 Calculate the variation of the internal energy of the system [(S), chute, Earth] between A and B.

Part 2 — The spring buffer

(S) reaches the spring at B and compresses it; the friction force still acts. At an instant, the compression of the spring is $X = x - AB$.

2.1 Write the expression of the mechanical energy of the system [(S), spring, Earth] in terms of m, g, α, k, X and v. By deriving with respect to time, show that the acceleration of (S) is $a = g\sin\alpha - \dfrac{f + kX}{m}$.

2.2 Show that the speed of (S) keeps increasing just after it touches the spring. Determine the compression for which its speed is maximum, and calculate this maximum speed.

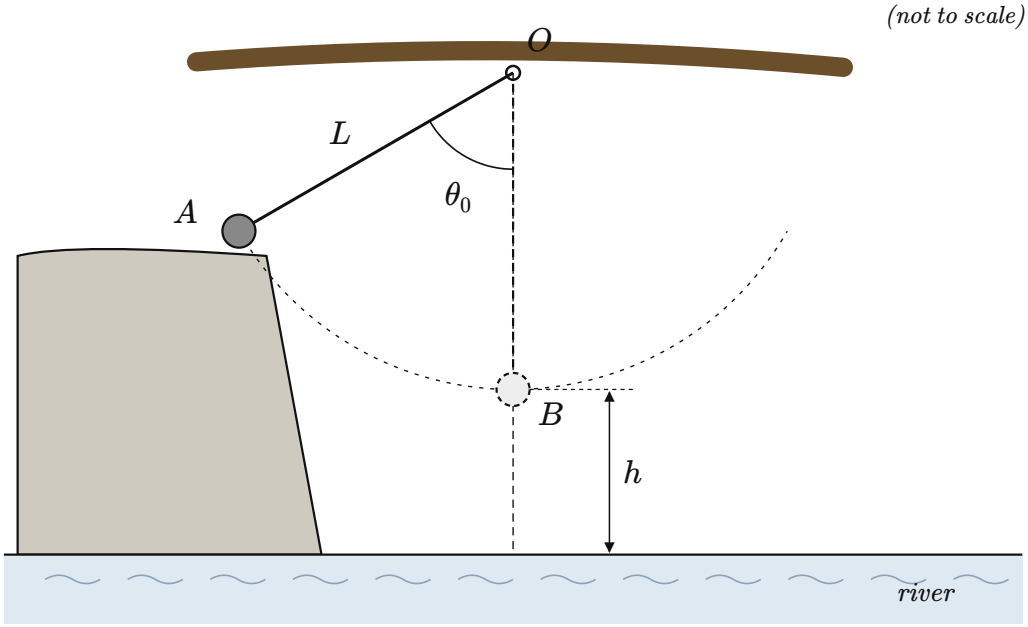
2.3 Show that the maximum compression of the spring is $X_m = 75$ cm.

2.4 Calculate the acceleration of (S) at the maximum compression. Describe the nature of the motion of (S) during the compression.

2.5 The maximum value of the friction force when (S) is at rest is also f. Show that (S) does not remain at rest at the maximum compression. Determine its speed when it leaves the spring, then the distance it travels back up the chute before it stops.

Exercise 2 — The rope swing over the river (10 points)

On a summer day at the river, a swimmer holds the end of a rope tied to a tree branch at O. Standing on a rock at A, the rope stretched and making the angle θ_0 with the vertical, he swings without initial velocity. At the lowest position B, he lets go of the rope and falls into the river (Doc. 3).



Doc. 3

> Mass of the swimmer: $m = 60$ kg; he is considered as a particle at the end of the rope.
> The rope is inextensible and of negligible mass; $OA = OB = L = 5.0$ m; $\theta_0 = 60°$.
> B is at the height $h = 3.2$ m above the surface of the water. Air resistance is neglected in Parts 1 and 2.
> Take: $\cos 60° = 0.50$; $\cos 34° \approx 0.83$; $\cos 48° \approx 0.67$; $\cos 56° \approx 0.56$.

Part 1 — The swing (from A to B)

The horizontal plane through B is the reference level for gravitational potential energy. At an instant, the rope makes the angle θ with the vertical and the swimmer has the speed v.

1.1 Name the forces acting on the swimmer. Justify that the work done by the tension of the rope is zero.

1.2 Applying the kinetic energy theorem between A and the position where the rope makes the angle θ, show that $v^2 = 2gL(\cos\theta - \cos\theta_0)$. Calculate the speed v_B at B.

1.3 Show that the mechanical energy of the system [swimmer, Earth] is conserved, and calculate its value.

1.4 Determine the angle θ_1 at which the kinetic energy is twice the gravitational potential energy, and the speed of the swimmer at this position.

1.5 Determine the angle θ_2 at which the gravitational potential energy is twice the kinetic energy.

Part 2 — The fall into the river

The swimmer lets go of the rope at B with the velocity $\vec{v}_B$, horizontal. The surface of the water is now taken as the reference level for gravitational potential energy.

2.1 Calculate the mechanical energy of the system [swimmer, Earth] at B with this new reference. Does the change of reference affect the speeds calculated in Part 1? Justify.

2.2 Determine the speed of the swimmer when he reaches the water.

2.3 Determine the height above the water at which his kinetic energy is twice the gravitational potential energy, and his speed at this height.

Part 3 — The real swing

Another day, the swimmer does not let go of the rope. Because of air resistance, he stops on the other side when the rope makes the angle 56° with the vertical. The length of the path he follows between A and this position is about 10 m.

3.1 Applying the kinetic energy theorem between A and this position, determine the work done by the air resistance.

3.2 Deduce the average magnitude of the air resistance and the variation of the internal energy of the system [swimmer, air, Earth].

— End of the test —